Richard Oliveira

Doctoral Researcher at Politecnico di Milano

Google Scholar LinkedIn GitHub Website

I'm a Marie Skłodowska-Curie Doctoral fellow working on distributed optimization and machine learning theory through the lens of applied harmonic analysis and signal processing.

Education

Politecnico di Milano

2025 - Present ▷ PhD in Data Analytics & Decision Sciences.
Advised by Dr. Stefano Savazzi and Prof. Monica Nicoli.

University of California, San Diego

2022 - 2024 ▷ MSc in Electrical Engineering, Signal & Image Processing.
Advised by Prof. Massimo Franceschetti.

2020 - 2022 ▷ BSc in Electrical Engineering.
Advised by Prof. Nick Antipa.

Research Experience

Politecnico di Milano
Electronics, Information Technology, and Bioengineering Department

2025 - Present ▷ Doctoral Researcher.
Advisor: Dr. Stefano Savazzi and Prof. Monica Nicoli.
Mathematical foundations of machine learning, focusing on consensus algorithms and implicit regularization.

University of California, San Diego
Electrical and Computer Engineering Department

2023 - 2024 ▷ Graduate Student Researcher.
Advisor: Prof. Massimo Franceschetti.
Contributed to the theory of discrete Pearson-Rayleigh random walks by analyzing how walks of this kind behave in discrete space with geometric and Poisson distributed step sizes.

2020 - 2022 ▷ Undergraduate Student Researcher.
Advisor: Prof. Nick Antipa.
Analysis and development of a linear, spatial-invariant imaging system, DiffuserCam; experimental recovery of a point spread function; Python implementation of an accelerated ADMM solver for inverse-problem scene reconstruction.

Publications

Journal Articles

Feb. 2026 ▷ On Solutions to Discrete-Time Inverse Problems via Sparse Superpositions of Decaying Shifted Heaviside Functions: A Hardy Space Approach
R. Oliveira and S. Savazzi. *IEEE Signal Processing Letters*.

Apr. 2025 ▷ Distributed Composite Regularization via Dual-Layer Consensus Optimization
R. Oliveira and S. Savazzi. *IEEE Transactions on Signal and Information Processing over Networks*.

July 2026 ▷ On the Stability of Consensus Gradient Dynamics for Regularized Distributed Deep Matrix Factorization: Explicit Bounds
R. Oliveira and S. Savazzi. *IEEE Signal Processing Letters*.

Manuscripts Under Review

June. 2026 ▷ Deep Neural Networks with Exponentially Damped Ridge Spline Activations: A Functional-Analytic Approach to Distributed Learning
R. Oliveira, S. Savazzi and M. Nicoli. Submitted to *IEEE Transactions on Neural Networks and Learning Systems*.

Projects

Nov. - Dec. 2024 ▷ **Inverse Problems in NeuroMANCER**
Used NeuroMANCER to solve an MNIST inverse problem with physics-informed constraints and non-smooth regularizers. Compared results against CVXPY and identified the careful fine-tuning required for non-differentiable regularization.

Mar. - Jun. 2024 ▷ **Deep Pose Estimation from 3D Point Clouds**
Developed a neural network for relative-pose prediction between 3D point-cloud sets from LiDAR scans and RGBD camera data from the EDEN dataset, addressing initialization sensitivity and the computational cost of repeated alignment in classical registration methods.

Jan. - Mar. 2024 ▹ **Visual-Inertial SLAM**
Implemented a SLAM pipeline using the Extended Kalman Filter: prediction for localization, updates for landmark mapping, and the combined prediction-update loop for simultaneous localization and mapping.

Jan. - Mar. 2024 ▹ **LiDAR-based SLAM**
Approached SLAM for a differential-drive robot using wheel encoders, IMU, LiDAR, and RGBD camera measurements. Used ICP and GTSAM for trajectory estimation and constructed an occupancy grid from the optimized trajectory and LiDAR measurements.

Jan. - Mar. 2023 ▹ **Theory and Applications of Convex Lifts**
Prepared a literature review on lifts of convex sets, emphasizing how higher-dimensional convex representations can simplify structure and support optimization methods.

Teaching

University of California, San Diego
Electrical and Computer Engineering Department

Mar. - Jun. 2024 ▹ Graduate Teaching Assistant, ECE 227: Big Network Data.
Developed robotics-oriented course projects for a course on graph analysis; provided feedback and instructional support during lecture hours.

Sep. - Dec. 2021 ▹ Undergraduate Teaching Assistant, ECE 35: Introduction to Analog Design.
Provided academic support during lectures, developed instructional materials, attended lab sessions, assisted in exam grading, and held weekly office hours.

Employment

Italian National Research Council (CNR)
Institute of Electronics, Computer and Telecommunication Engineering (IEIIT), Milano

2025 - Present ▹ Doctoral Researcher.
Distributed optimization theory, regularization, and consensus networks.

Onikron LLC
Remote

2024 - 2025 ▹ Junior Data Scientist.
Implemented and researched modern NLP models, including transformers and RNNs in PyTorch; developed lemmatization and tokenization scripts for data preprocessing; worked on clustering for unsupervised learning.

General Atomics A.S.I.
San Diego, CA

Jun. - Aug. 2023 ▹ Algorithms and Signal Processing Engineer.
Researched modern neural-network architectures for computer-vision deep learning; created a Python/Bokeh data-visualization tool for heat-map analysis and preprocessing.

Ciena
San Jose, CA

May - Aug. 2021 ▹ Database Engineer.
Used SQL queries through the FLYBID database to organize sensitive company records; focused on data dictionaries and LucidChart-based data mapping.

Selected Honors & Awards

2025 - 2028 ▹ Marie Skłodowska-Curie Doctoral Fellowship.
Jan. - Mar. 2024 ▹ Graduate Student Researcher Grant.

Skills

Programming ▹ C, C++, MATLAB, SQL, Python, PyTorch, TensorFlow, Assembly, Scikit-learn.
Languages ▹ Italian; English.